Ice Hunters

Team 24 Participants


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Agenda

Part 1	Part 2	Part 3	Part 4
Mission & Science Objectives	Engineering / Payload	ConOps / Rover Specifics	Schedule & Budget

Mission Overview

Goals

To assess a lunar water-ice strategy for science investigation on the South Pole of the Moon

Mission Objectives

- Selecting appropriate science instruments to conduct investigation
- Design a spacecraft capable of operating on the surface of the Moon
- Establish programmatics plan for management, costs, and schedules

Science Objectives

Science Objective 1:
- Drill 100g of ice samples at one of the 13 Artemis III landing site locations and determine the composition of the sample.
- NSS

Science Objective 2:
- Determine the water ice abundance of at least one of the 13 Artemis III landing site locations near or within a PSR within the top 1 meter of regolith.
- NIRVSS

Science Objective 3:
- Identify areas of higher ice concentration within an Artemis III landing site.
- NSS and NIRVSS

The Importance / Science Goals

In Accordance with the Artemis III Science Goals…

Objective 2: *Understanding the character and origin of polar volatiles*

Primary: 2c. Understand the transport, retention, alteration, and loss processes that operate on volatile materials near and within permanently shadowed regions.

Secondary: 2a. Determine the compositional state (elemental, isotopic, mineralogic) and compositional distribution (lateral and with depth) of the volatile component in lunar polar regions.

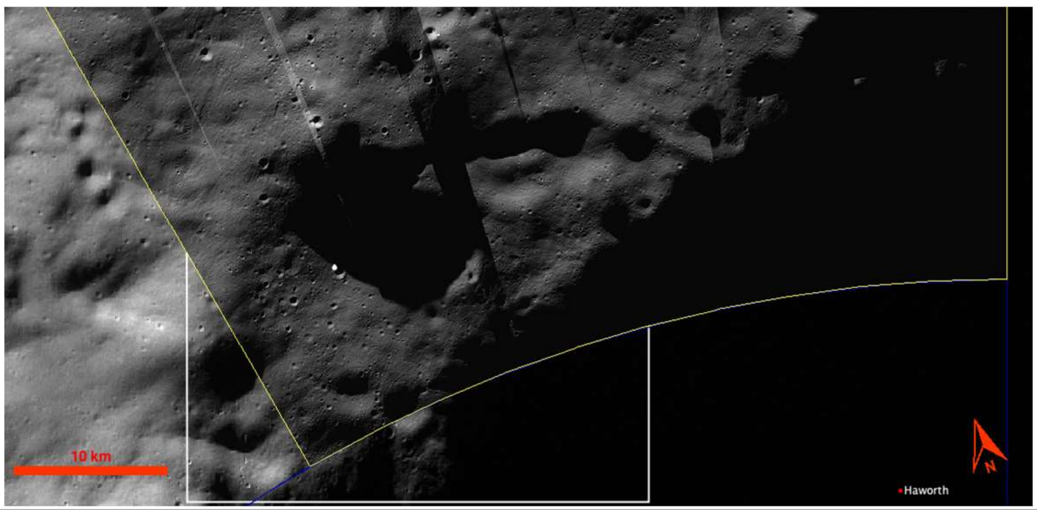
4. (NASA, "Artemis III Science.").

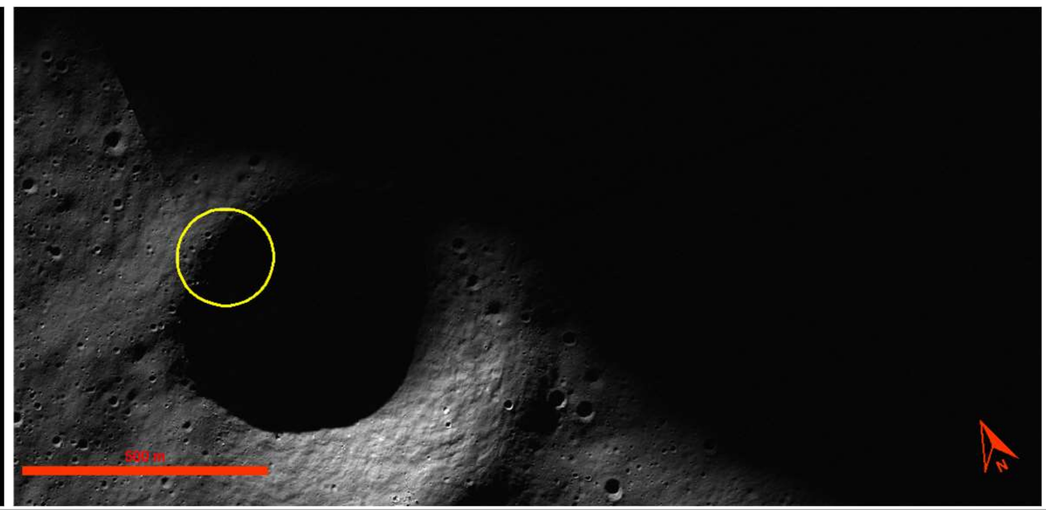
Location - Haworth

Region selection - Spacecrafts such as the Clementine, Lunar Prospector, and Lunar Reconnaissance Orbiter have provided evidence of water-ice

Selection criteria - Hydrogen abundances, sunlight availability, low slopes

Images courtesy of JMARS (-86.721°N,





Mission Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
0.1	The system shall survive the lunar environment for 2 years.	In order to achieve science objectives the system needs to operate/survive	Client	SYS-03	Demonstration	All	Met
0.2	Shall investigate the abundance of near-surface water ice in PSR	Foundational science goal	Client	SYS-01.SYS-03,SYS-05,SYS-06,SYS-07,SYS-09	Test	Payload	Met
SYS-01	The system shall traverse the lunar surface by means of a rover	The rover must be able to travel distances to conduct science objectives	0.1,0.2	SYS-07,SYS-09	Demonstration	Mechanical, Electrical, GNC, Computer hardware	Met
SYS-02	The system shall not exceed a mass of 85 kg	Defined by launch provider	Launch provider	All	Inspection	All	Met
	The system shall be able to operate at temperatures ranging	The rover must function in extreme temperatures of the			Demonstration	Electrical, Thermal,	Met

Mission Requirements

SYS-04	The system shall not exceed a volume of L 1.5m x W 1.5m x H 1.5m	Defined by launch provider	Launch provider	All	Inspection	Structural	Met
SYS-05	The system shall collect and store data from the two science instruments on the rover	The data collected will contribute to achieving the science goal	0.1,0.2	SYS-01,SYS-03,SYS-07	Demonstration	Science, Electrical	Met
SYS-06	The system shall communicate with the LRO	After data has been collected and analyzed it will be relayed to earth	0.1,0.2	SYS-07	Demonstration	Electrical, Computer hardware, GNC	Met
SYS-07	The system shall have sufficient power to survive/operate	The rover requires power to perform and communicate all objectives	0.1,0.2	SYS-03	Demonstration	Electrical	Met
SYS-08	The system shall not exceed a cost of $225M	Defined by contracting client	Contract provider	All	Verification	All	Met
SYS-09	The system shall contain the necessary instruments to support the collection from the scientific instruments	The rover needs the ability to run experiments on the Moon's surface	0.2	SYS-05,SYS-07	Demonstration	Science, Electrical, Computer hardware, Thermal	Met

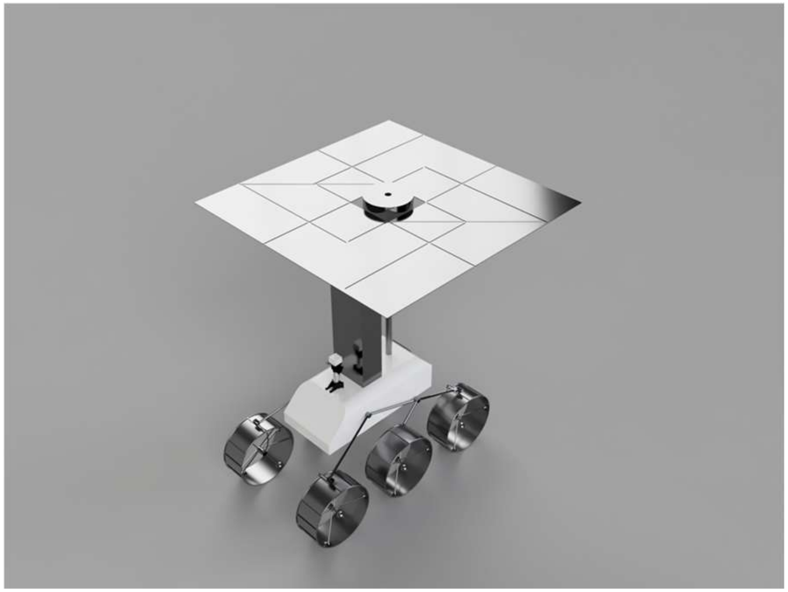
Mission Requirements

SYS-10	The system shall not contain a Radioisotope Thermoelectric Generator(RTG)	The rover cannot have any derivative of this concept	Contract provider	All	Verification	All	Met
SYS-11	The system shall be ready for launch by September 1st, 2028	In order to align with the launch provider's secondary payload	Contract provider	All	Verification	All	Met
SYS-12	The system shall not contain radioactive material exceeding a mass of 5g	Limited radioactive materials	Contract provider	All	Verification	All	Met

Vehicle Design

Mass (kg)	Volume (cm^3)	Power (W)
72.61	502,463	432

- Mid-Size Rover
- Rocker-Bogie Suspension
- Deployable Solar Panel
- NSS & NIRVSS Instruments
- Drill



Mechanical

Components:

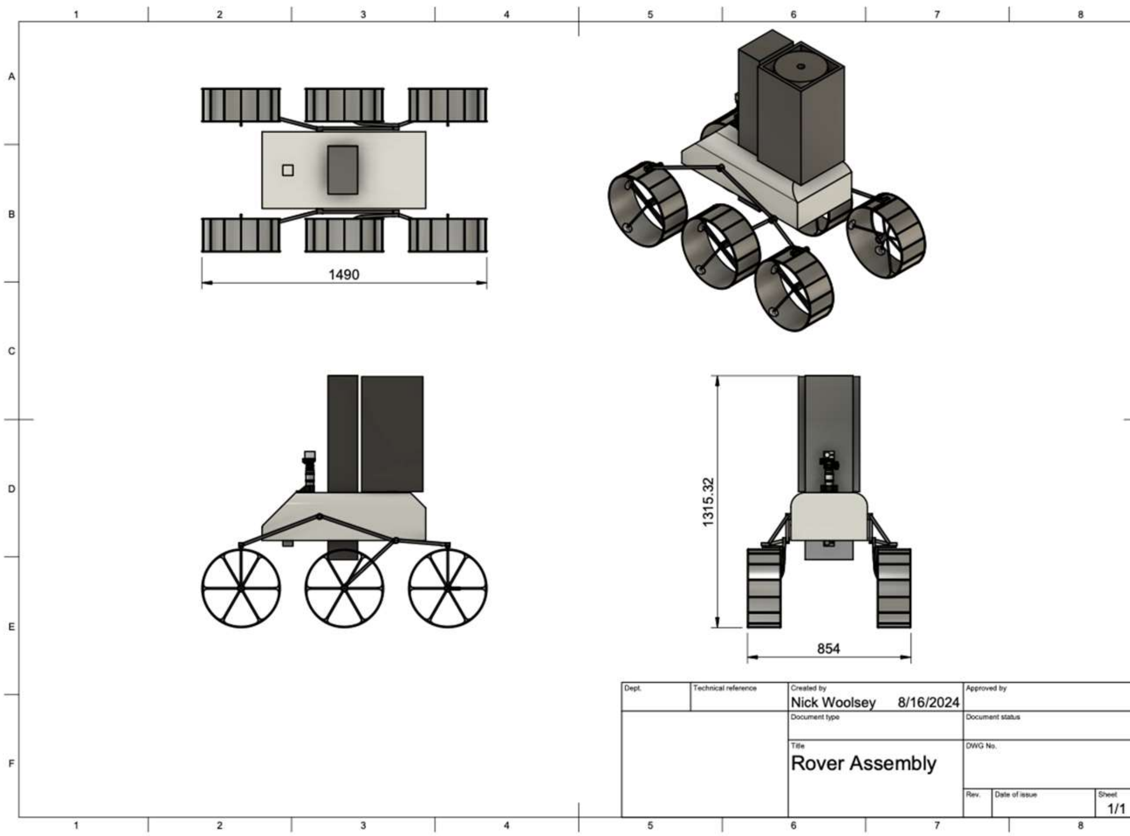
- Materials
  - Aluminum 2000 and 6000 series
- Mobility system
  - Six wheeled rocker-bogie system
- Collection devices
  - Drill
  - Energy
  - Imaging

Component	Mass[kg]	Volume[m]	Power draw [W]	TRL
Chassis	15	1x.5x.25	10	7
Suspension	1	.8x.3x.05	2	7
Wheels	6	.349	24	7
Drill	12	.145x.025x.126	3	6

Mechanical

Features:

- ~11'' diameter wheels capable of overcoming obstacles two times their height
- Low complexity suspension system
- High strength low mass material



Power
Requirements

Requirement Number	Requirement Description	Rationale	Parent Req	Requirement met?
PS-01	The Power Subsystem shall be able to withstand the temperature range that the rover experiences	The system must be able to work in these operating conditions on the moon.	SYS-03	Yes Verified by datasheet
PS-02	The Power Subsystem shall be able to withstand the environmental radiation on the surface of the moon. (1,369 microsieverts per day)	The system must be able to function on the surface of the moon for the lifetime of the mission	PS-06 PS-07	Yes Verified by datasheet
PS-03	The Power Subsystem shall be able to supply all power requirements of the spacecraft between solar charging intervals.	The system must be able to ensure that the science goals of the craft can be fulfilled and therefore must be able to supply all instruments with power for up to 1 week	SYS-07	Yes Verified by calculation
PS-04/PS-05	The Power Subsystem shall take up no more than 20 percent of the total allotted mass and volume. (17.1 kg)(0.675m^3)	The system must not encroach on the maximum weight and volume requirements for the whole craft	SYS-02	PS-04 Excepted PS-05 Met

Power

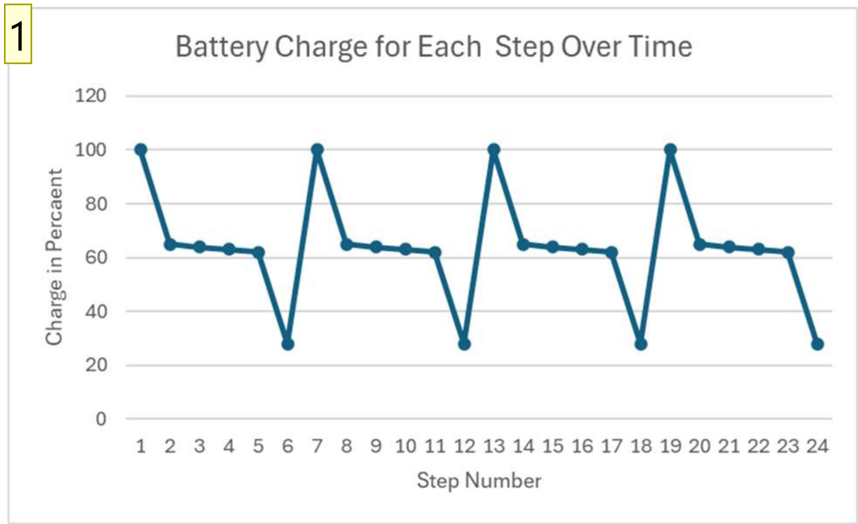
Components

- ROCKETLAB ZTJ Triple Junction solar cell
- Eaglepicher High-Energy 55Ah Lithium Ion Cells
- AAC Clyde Space STARBUCK-MINI 1500w Power supply
- System weight and volume

Part Name	Weight (kilograms)	Volume (centimeters)	Power Handling Requirements (watts)	Power Capabilities (watts)
Solar Cells	1.4 kg	150 x 150 x 2.2 cm total volume	Supply 400W	400W
Battery Array	20.01 kg	0.22 meters^3	Supply 450Wh	610Wh
Power Supply	3.3 kg	21.5 x 11.5 x 5.0 cm	432W max	1500W
Total:	24.7 kg	226288.75 cm^3 (0.6m^3)	N/A	N/A

Power

1



Step 1: Surface deployment and traverse from crater rim to test location 1
Step 2: NSS and NIRVSS data collection at location 1
Step 3: TRIDENT drill sample collection
Step 4: Data processing
Step 5: Data transfer from rover to LRO
Step 6: Traverse to another test site or crater rim and repeat

2. NASA LSPACE Power vs Time Graph

Slide 16

1 cite

Matthew Mulderink, 8/17/2024

Command and Data Handling

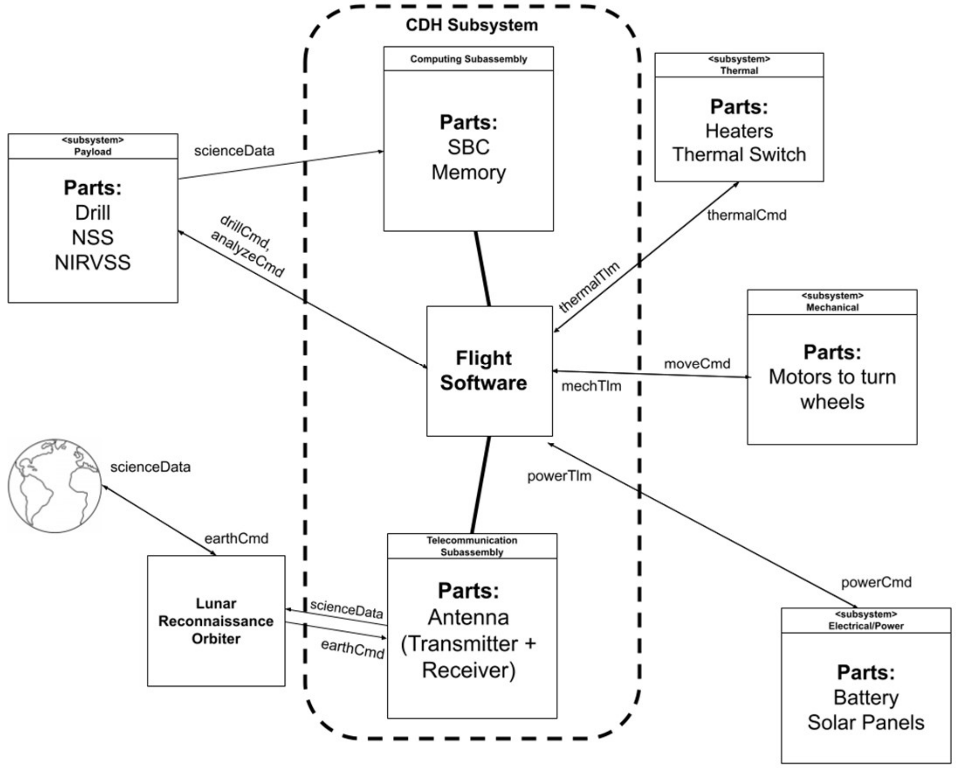
Components:

- Single-Board Computer: BAE Systems RAD5545™
- Data Bus: SpaceWire
- Antenna (Transmitter + Receiver): BAE Systems Ka-band Multi Orbit LEO Optimized Antenna
- Memory: 2GB Flash Storage
- Software: Linux Operating System

Component	Mass (kg)	Volume (L*W*H) (cm)	Power Requirements (watts)	TRL
Single Board Computer	0.24 kg	18.5 cm x 18 cm x 4.5 cm	30W max	5
Transmitter	7kg T+R	2 Transmit Subarrays: 20 cm x 40 cm	55W upload when sending data to LRO	6
Receiver		6 Receive Subarrays: 45 cm x 34 cm	65W download when receiving data from LRO	6
Data Bus	0.03 kg/m	N/A	N/A	7
Storage	0.03 kg	5 x 2.5 x 1 cm^3	Write: 0.0067 W Read: 0.0027 W Off: 0 W	6
Software	N/A	N/A	Low	7
Total:	7.57 kg	3507.8 cm^3	150.0094 W	5

Command and Data Handling

Software Architecture Flowchart:



Thermal

- Passive System
  - Sierra Nevada Thin Plate Heat Switch
  - Sprayable Thermal Control Coating (Socomore A276 Polyurethane)
  - Q-Rad Deployable Radiator
- Active System
  - 8 Omega's KHLVA Polyimide Heater

PLM-Series

Lockheed Martin Space's MICRO1-1

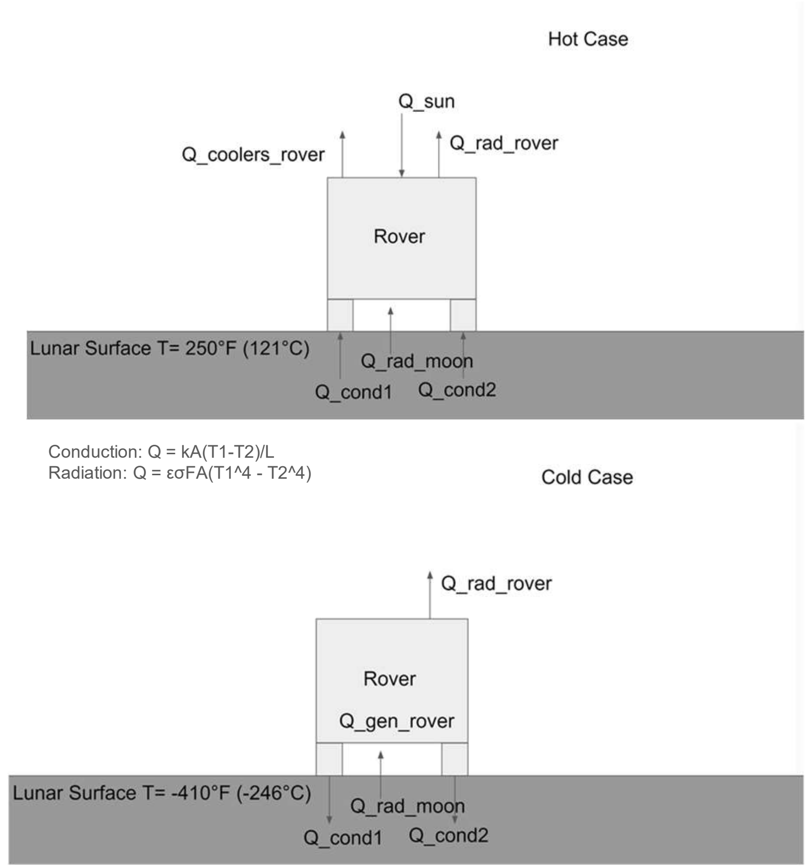
Cryocooler

Component	Mass (kg)	Volume	Max Power	TRL
Heat Switch	0.032	1 x 1 x 0.32 in^3 (25.4 x 25.4 x 8.13 mm^3	12 W	7
Sprayable Coating	0.003	1 gallon (The entire bucket will not be used)	N/A	6
Radiator	0.098	10 cm x 20 cm x 3 cm	N/A	7
Heater	N/A	18 mm x 15 mm x 3.6 mm	0.4 to 1.56 (W/cm^3)	7
Cryocooler	0.350	N/A	15 W	9

Thermal

Hot Case
- Surface T = 394 K
- Q_coolers_rover = 116.66 W

Cold Case
- Surface T = 27 K
- Q_gen_rover = 100.87 W



Payload

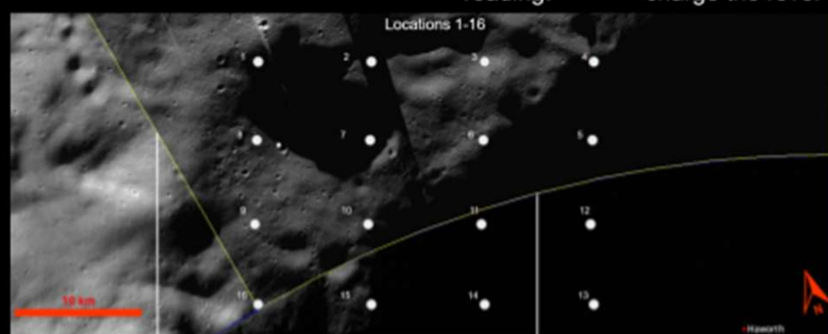
- Neutron Spectrometer System (NSS)
  - Measures thermal flux to determine H%

- Near Infrared Volatile Spectrometer System (NIRVSS)
  - Measures surface water
  - Uses LEDs to measure 340-940 nm
  - Monitors reflectance and temperature

Component	Mass (kg)	Volume (L*W*H) (cm^3)	Power Requirements (Watts)	TRL
NSS	1.6	Sensor Module: 21.3 x 32.1 x 6.8 cm Data Processing Module: 13.9 x 18.0 x 3.0 cm	1.5	6
NIRVSS	3.57	18.0 x 17.85 x 8.55	30	6

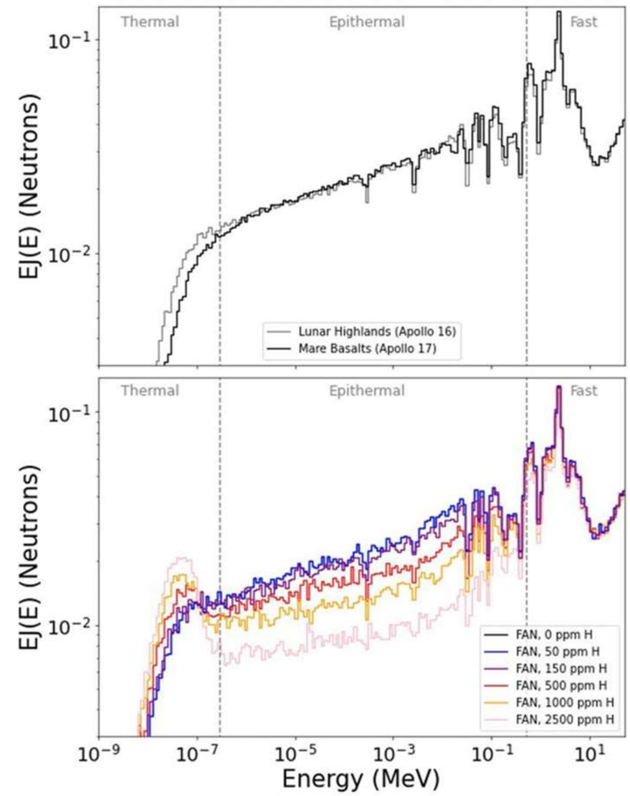
L'SPACE Team 24 "Ice Hunters" Rover Concept of Operations

T+1 Day	T+2 - 25 Days	T +26 Days	T +27-29 Days	T +30 Days	T +31-480 Days
Surface Deployment Rover land at sites. Systems Check NIRVSS & NSS check Comms, Power, Mechanical and Thermal check Thermal System maintains operating temperature in hot & cold cases	Driving Mode (Day) Drive to Location 1 at 0.8 mph. Solar panels deployed to charge rover Driving Mode (Night) Drive at 0.4 mph.	Science Mode Obj 2 Arrive at Location 1 and turn on NIRVSS. Measure wavelengths of 250-600 nm to determine ice water abundance	Science Mode Obj 1 Turn on NSS and take spectroscopy reading of 3x3cm^2 area. Drill the 3x3cm^2 and collect 100g of lunar regolith. Use NSS to take spectroscopy reading.	Data Transfer Mode NSS and NIRVSS in standby mode Send data from science instruments Charging Mode When exposed to light, solar panels charge the rover	Drive to Locations 2-16 and repeat Science Mode Obj 1, Science Mode Obj 1, Data Transfer, and Charging Mode.



Expected Data & Analysis

- NSS
  - Cd wrapped neutron sensors
  - Neutrons reflect energy differently based on the composition
  - Neutron spectra used to determine H%

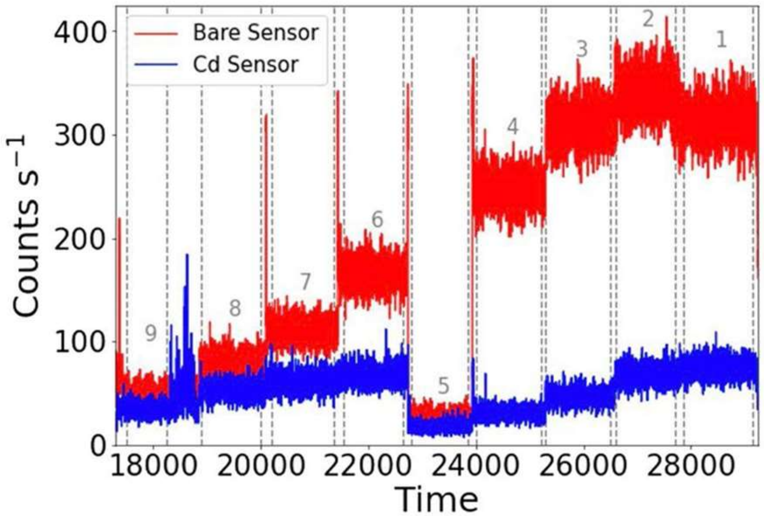


Neutron Spectra Graphs[2]

2. Peplowski, P. N. Calibration of Nasa's Neutron Spectrometer System

Expected Data & Analysis

- NSS
  - Counts vs Time graph correlates to H abundance
  - Science Objective 1
    - Compare H% before and after drilling
    - High Count = H lacking region
    - Low Count = H abundance

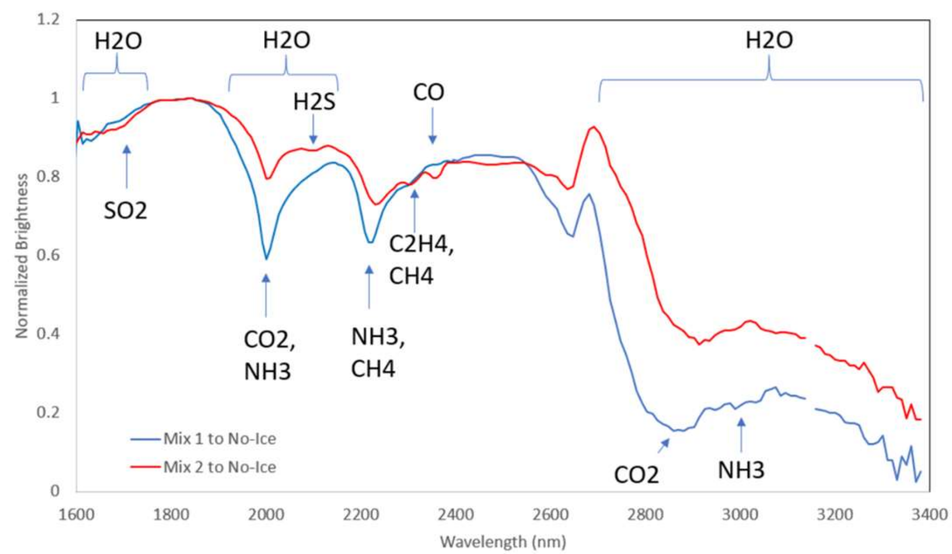


Time-series data of the bare- and Cd-covered sensors for each pile configuration[2]

2. Peplowski, P. N. Calibration of Nasa's Neutron Spectrometer System

Expected Data & Analysis

- NIRVSS
  - Uses LEDs to collect absorption, emission, and reflectance data
  - Science Objective 2
    - Detect H wavelengths of 400-680 nm in the top 1 meter of regolith
  - Science Objective 3
    - Determine 3 locations with the highest %H
    - Temperature and reflectance data provides context



Initial NIRVSS Spectral Data on the two versions of OPRFLCROSS icy regolith simulant[3]

3. Roux, Vincent Testing NIRVSS

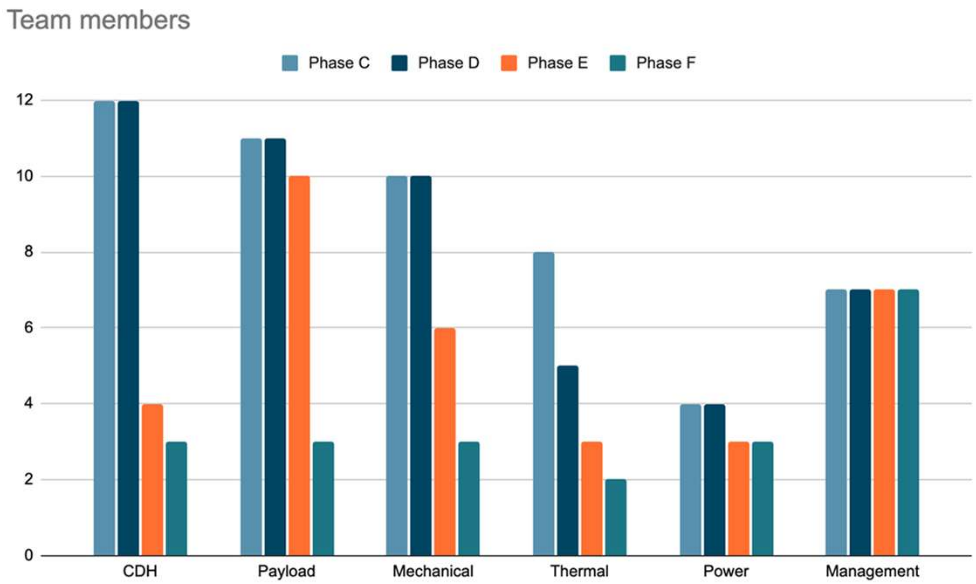
N^2 Diagram

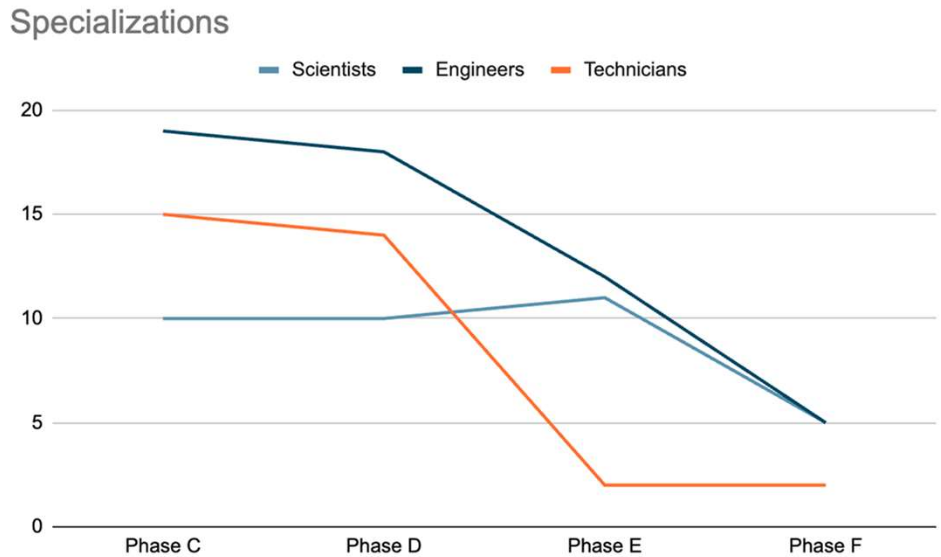
Payload	↴instrument info				
⤶ instrument commands	**Data Handling**	↴ instrument data, rover telemetry	↴ location to go to	↴ needs of subsystems temperature	↴ when to get more power
	⤶ comms to rover	**Communications with LRO**			
	⤶ telemetry		**Navigation and Motion Control**		
⤶ when samples are collected, specific thermal	⤶ current temperature			**Thermal**	
⤶ Power for instruments	⤶ Power for computation and processing	⤶ Power for transceiver	⤶ Power for motors and attitude determination sensors	⤶ Power for sensors and heating elements	**Power**

Risk Matrix

Function	Failure Mode	Effects	Cause	Prevention	Actions
Power	Solar cells fail to charge batteries.	System loss of power resulting in mission failure.	Incorrect timing of the rover's charging could leave it stranded in a shadowed region without the ability to recharge via solar panels.	Managing battery usage for data collection, driving, and system power, while planning routes with charging stops and time margins.	If this failure continues, the number of science locations can decrease and increase the number of charging locations. Redundant batteries can be used, however, they carry extra mass.
Mechanical	Solar arrays fail to deploy.	System loss of power resulting in mission failure.	Onboard mechanical systems fail to deploy correctly causing the solar arrays to be unusable.	Testing deployment procedure in a Moon-like environment and ensuring robustness.	Implementing redundancy into the system to ensure the rover has some level of recharge capabilities.
CDH	Radiation exposure leading to data and equipment malfunction.	Could permanently damage electronics or cause data interference/loss.	Ineffective radiation shielding.	Use a test bed to verify radiation shielding and add extra protection around the rover's storage drive.	Back up data into a secondary drive. Add extra radiation shielding to the hard drive. Coordinate data collection timing with the satellite's data transfer range timing to reduce the time the data is stored.
Mechanical	Wheels of the rover wear over time in lunar terrain.	Science objective 3 cannot be accomplished if the rover is unable to travel to multiple locations.	Using a testing environment that does not accurately represent the terrain and environment of the Moon. Not being able to test the wheels for 3 years.	Use wheel design and material with heritage, especially 3+ years on the Moon.	Reduce the number of science locations. Increase the number of wheels (most likely not possible due to mass). Increase wheel durability.
Software	Critical software fails leading to system malfunction.	Could eliminate the use of critical components or the rover as a whole.	Software bugs or unexpected scenarios that were not properly accounted for.	Ensure code redundancy for rover software robustness and conduct broad testing to cover all possible scenarios.	Critical failure reset and bug awareness procedures.

Project Management Overview





Overview of Schedule

Phase C: 10/1/24 to 3/31/26
Phase D: 4/1/26 to 9/30/28
Phase E: 9/1/28 to 9/1/29
Phase F: 9/1/29 to 9/1/30

Milestone	Date
Full-scale assembly of rover instruments and subsystems complete	5/9/25
Completion of Initial Component Orders	7/16/25
Completion of Fabrication for Test	12/18/25
Systems Integration Review (SIR)	8/28/26
Critical Design Review (CDR)	9/30/26
Subsystem Testing Complete	6/11/27
Integrated Testing Complete	9/17/27
Launch Readiness Review (LRR)	12/17/27
Launch Date	9/1/28

Budget Overview

FINAL COST CALCULATIONS							
Total F&A	$ 1,850,000	$ 2,643,502	$ 1,434,747	$ 1,881,839	$ -	$ -	**$ 7,810,088**
Total Projected Cost	$ 33,161,926	$ 52,970,986	$ 27,925,710	$ 39,497,979	$ 5,423,152	$ 3,153,895	**$ 162,133,649**
Total Cost Margin	$ 2,707,472	$ 4,316,644	$ 2,141,690	$ 3,005,913	$ 330,177	$ 181,969	**$ 12,683,865**
	8.2%	8.1%	7.7%	7.6%	6.1%	5.8%	
Total Project Cost	$ 33,161,926	$ 52,970,986	$ 27,925,710	$ 39,497,979	$ 5,423,152	$ 3,153,895	**$ 162,133,649**

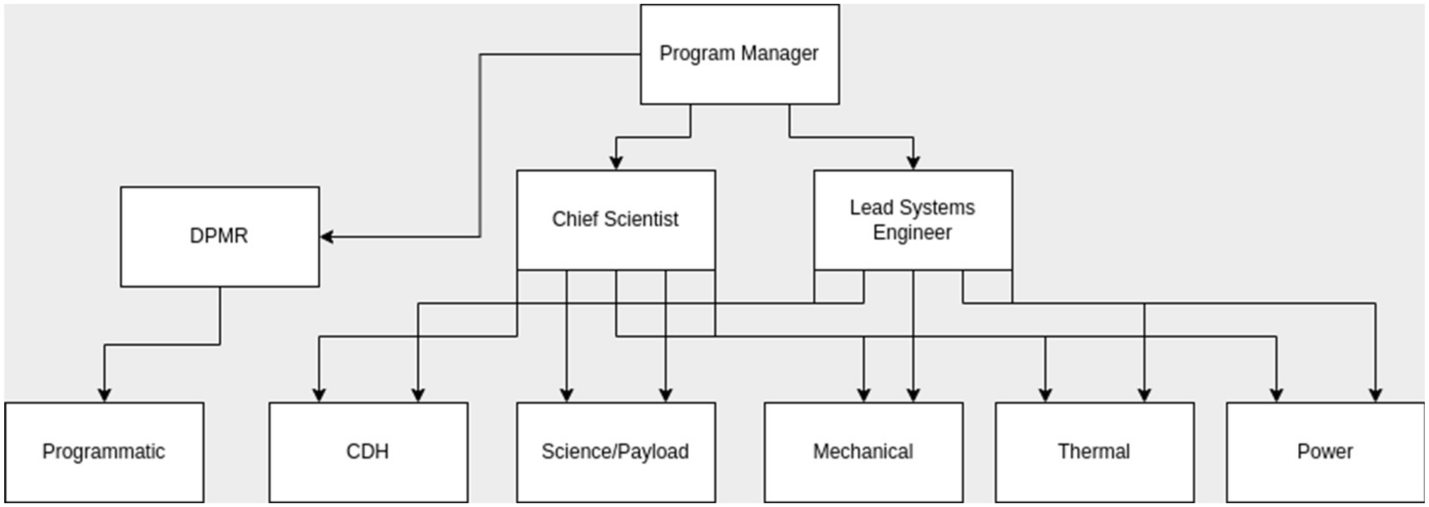
Budget Basis of Estimates

- ❑ Complexity of mission
- ❑ Margins
- ❑ Travel rates
- ❑ Manufacturing facility costs
- ❑ Testing facility costs
- ❑ Outreach
- ❑ Inflation
- ❑ Direct Costs

Personnel Budget and Team Management

	Phase C	Phase C	Phase C-D	Phase D	Phase E	Phase F
# People on Team	FY 1	FY 2	FY 3	FY 4	FY 5	FY 6
Science Personnel:	10	10	10	10	11	5
Engineering Personnel:	19	19	18	18	12	5
Technicians:	15	15	14	14	2	2
Administration Personnel:	2	2	2	2	2	2
Management Personnel:	5	5	5	5	5	5

Budget: $30,205,108 (18.3%)



Travel Budget

Total cost: $816,534 (0.5%)

Travel Plans:

- FY1 and FY2: 4 design review trips each (5 members, 2 nights)
- FY3: 6 component testing/fabrication trips (7 members, 3 nights)
- FY4: 8 fabrication and integration trips (10 members, 5 nights)
- FY5: 1 launch trip (32 members, 5 nights)
- FY6: 5 operations trips (10 members, 4 nights)
- Assumptions: CDR and SIR don't necessarily require travel
- Flexibility: Morale trips, in-person reviews, operations

Direct Costs

DIRECT COSTS							
Mechanical Subsystem	$ 5,000,000	$ 9,000,000	$ 6,000,000	$ 6,000,000	$ -	$ -	**$ 26,000,000**
Power Subsystem	$ 500,000	$ 700,000	$ 824,000	$ 607,200	$ -	$ -	**$ 2,631,200**
Thermal Control Subsystem	$ 1,000,000	$ 1,000,000	$ 747,200	$ 824,160	$ -	$ -	**$ 3,571,360**
Comms & Data Handling Subsystem	$ 6,000,000	$ 9,000,000	$ 3,000,000	$ 5,400,000	$ -	$ -	**$ 23,400,000**
Science Instrumentation	$ 6,000,000	$ 6,000,000	$ 3,000,000	$ 4,500,000	$ -	$ -	**$ 19,500,000**
Spacecraft Cost Margin	$ 1,850,000	$ 2,570,000	$ 1,357,120	$ 1,733,136	$ -	$ -	**$ 7,510,256**
Total Spacecraft Direct Costs	$ 20,350,000	$ 29,005,020	$ 15,704,593	$ 20,551,527	$ -	$ -	**$ 85,611,139**
Manufacturing Facility Cost	$ 5,000,000	$ 13,800,000	$ 2,600,000	$ -	$ -	$ -	**$ 21,400,000**
Test Facility Cost	$ -	$ -	$ 1,500,000	$ 8,592,000	$ -	$ -	**$ 10,092,000**
Facility Cost Margin	$ 500,000	$ 1,380,000	$ 410,000	$ 859,200	$ -	$ -	**$ 3,149,200**
Total Facilities Costs	$ 5,500,000	$ 15,574,680	$ 4,744,520	$ 10,188,394	$ -	$ -	**$ 36,007,594**

Direct Costs Basis of Estimates

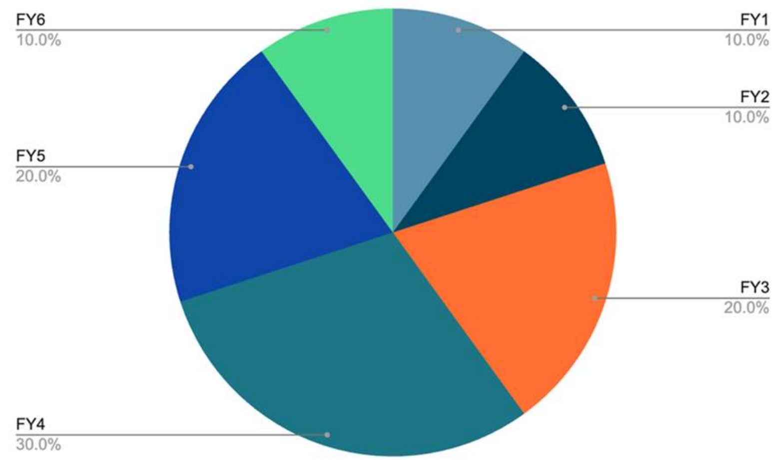
- ❑ Margin
- ❑ Ordering and procurement
- ❑ Facilities cost and timeline
- ❑ LLM use cases

Outreach Budget

OUTREACH							
Total Outreach Materials	$ 15,000	$ 15,000	$ 30,000	$ 100,000	$ 75,000	$ 50,000	**$ 285,000**
Total Outreach Venue Costs	$ 3,000	$ 3,000	$ 30,000	$ 150,000	$ 100,000	$ 40,000	**$ 326,000**
Total Outreach Travel Costs	$ 3,000	$ 3,000	$ 30,000	$ 150,000	$ 100,000	$ 40,000	**$ 326,000**
Total Outreach Services Costs	$ 8,000	$ 8,000	$ 50,000	$ 100,000	$ 100,000	$ 25,000	**$ 291,000**
Total Outreach Personnel Costs	$ 15,000	$ 15,000	$ 60,000	$ 75,000	$ 60,000	$ 25,000	**$ 250,000**
Outreach Margin	$ 2,200	$ 2,200	$ 10,000	$ 28,750	$ 21,750	$ 9,000	**$ 73,900**
Total Outreach Costs	$ 46,200	$ 47,401	$ 220,920	$ 650,843	$ 504,252	$ 213,570	**$ 1,683,186**

Outreach Strategies

- News Outreach: 20%
- Digital Media Outreach: 50%
- In-person outreach: 30%



Outreach Plan

- News
- Digital Outreach
  - Social Media
  - Videos
  - Citizen Science
  - K-12 Lessons and Activities
- In Person Outreach
  - Demonstration
  - Internships
- Partner Outreach

Social Media Posting Schedule		
Platform	**Frequency**	**Media Type**
Facebook	Weekly	Photos, Infographics, Short Videos
Instagram	Weekly	Photos, Infographics, Short Videos
X	Weekly	Photos, Infographics, Short Videos
Youtube	Bi-Monthly	Short Videos, Long videos

Ice Hunters PM Overview

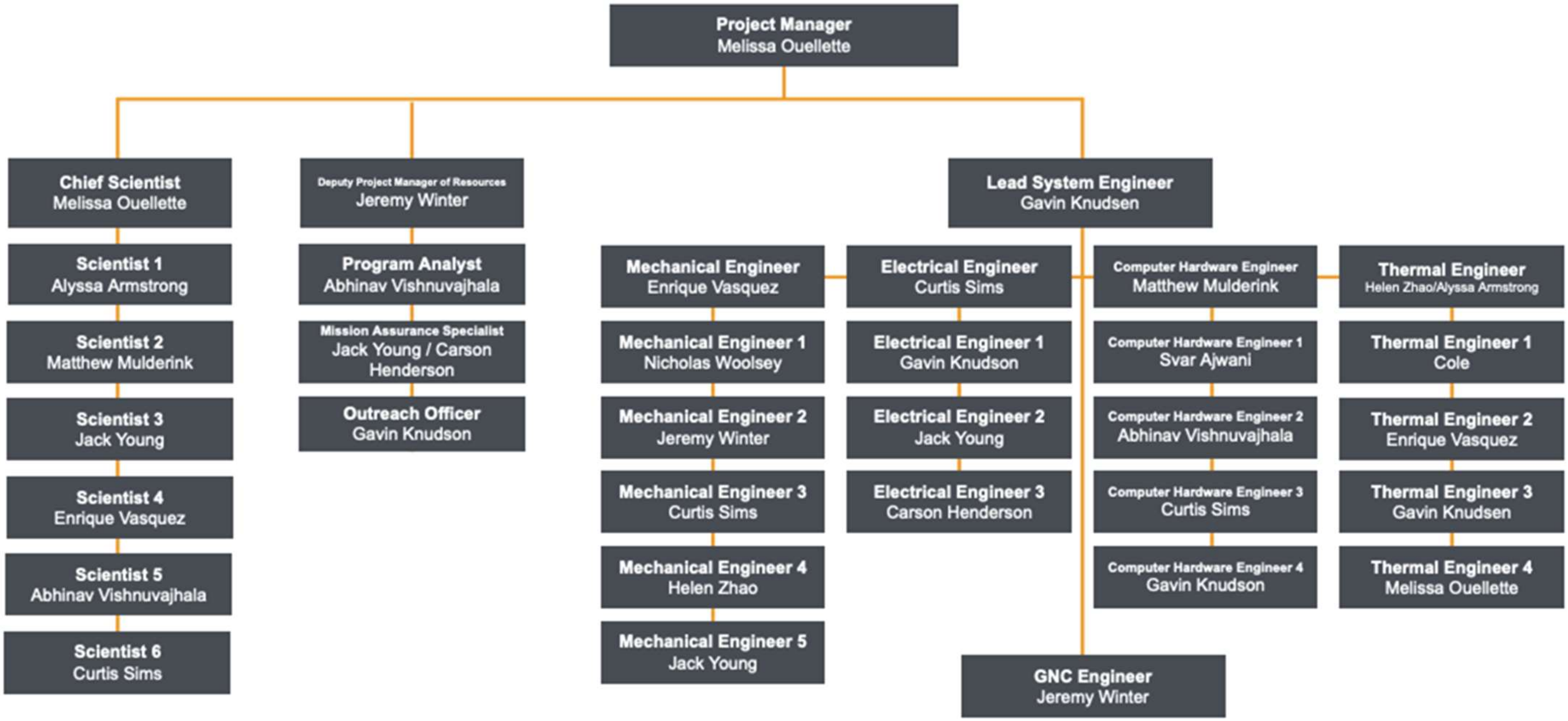
Challenges

- Lost 2 team members, including original PM who was really good
- Needed extensions to recover…
  - SRR: 5 days
  - MDR: 2 days
- Redo structure after SRR, refocused
- Meant that certain people could thrive, a few parts struggled
  - Could only stretch those people so thin to cover

Complete Org Chart

- **Project Manager**
  Matthew Mulderink
  - **Scientist**
    Alyssa Armstrong
    Jack Young
    Enrique Vasquez
  - **Deputy Project Manager of Resources**
    Jeremy Winter
    - **Program Analyst**
      Abhinav Vishnuvajhala
      Secondary (ish):
      Jack Young
      Svar Ajwani
    - **Mission Assurance Specialist**
      Nicholas Woolsey
      Secondary:
      Helen Zhao
    - **Outreach Officer**
      Gavin Knudsen
  - **Lead System Engineer**
    Gavin Knudsen
    - **Mechanical Engineer**
      Enrique Vasquez
      Secondary:
      Curtis Sims
      Nicholas Woolsey
    - **Electrical Engineer**
      Curtis Sims
    - **Computer Hardware Engineer**
      Svar Ajwani
      Secondary:
      Matthew Mulderink
    - **Thermal Engineer**
      Alyssa Armstrong
      Helen Zhao

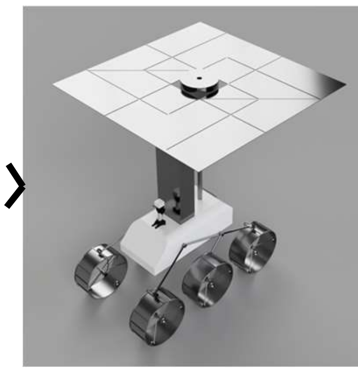
Original Roles



Summary

1. Objectives
- To assess Moon's PSR for Ice

2. Engineering
- NIRVSS
- NSS
- Engineering to support



3. Mission Specifics
- Expected Data
- 480 day ConOps
- N^2 & Risk

4. Schedule & Budget
- Timeline
- Team Management
- Budget

Next Steps

PDR Approval

↓

Phase C Begins

↓

Create hardware & software	Update Risk & Monitor Progress	Create plan for launch site

↓

Complete Reviews (in order): CDR/PRR, SIR, Safety Review

↓

Phase C Complete: To Phase D

1. NASA, "Program/Project Life Cycle."

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1. NASA. "Program/Project Life Cycle," February 6, 2019. Accessed August 17, 2024. https://www.nasa.gov/reference/3-0-nasa-program-project-life-cycle/.
1. NASA. "7.0 Thermal Control" February 24, 2024. Accessed August 17, 2024. https://www.nasa.gov/smallsat-institute/sst-soa/thermal-control/#7.1
2. Peplowski, P. N., R. C. Elphic, E. L. Fritzler, and J. T. Wilson. "Calibration of Nasa's Neutron Spectrometer System (NSS) for landed measurements of hydrogen content of the lunar subsurface." Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment 1049 (2023): 168063.
3. Roux, Vincent, Melissa Roth, Lauren Roux, Amanda Cook, and Anthony Colaprete. Testing NIRVSS with OPRFLCROSS Lunar Icy Regolith Simulants. Accessed August 6, 2024. https://www.hou.usra.edu/meetings/leag2019/eposter/5028.pdf.
4. Artemis III - Science Definition Report. Accessed August 17, 2024. https://www.nasa.gov/wp-content/uploads/2015/01/artemis-iii-science-definition-report-12042020c.pdf.

Questions?

Ice Hunters

Thermal Requirements

Requirements

- The system shall maintain an internal temperature of 20 ± 15 degrees Celsius at all times until the end of the mission.
- The system shall maintain the defined internal temperature range when at max and minimum power.
- The system shall maintain the defined internal temperature range in the coldest and hottest lunar surface temperatures (90.15 K to 379.15 K)
- The system shall not cost greater than the amount of money allocated to the thermal system.
- The system shall have a thermal model that shows net heat transfer equal to zero.
- The thermal model shall include the main forms of heat transfer in space, including conduction and radiation.
- The thermal diagram shall model extreme hot and cold cases.
- Exposed exterior components such as solar panels shall survive extreme lunar surface temperatures from 90.15 K to 379.15 K throughout the duration of the mission.
- The internal thermal system shall not exceed internal structural dimensions.

Thermal Equations

- Conduction: Q = kA(T1-T2)/L
- Radiation: Q = εσFA(T1^4 - T2^4)
- In the Hot Case:
    - Q_net in = Q_net out
    - Q_sun + Q_cond1 + Q_cond2 +Q_rad_moon = Q_rad_rover + Q_coolers_rover
    - Qcond1+2 = 89.56 W
    - Q_rad_moon = about 0 W
    - Q_rad_rover = 440.4 W
    - Q_sun = 467.5 W
    - Q_coolers_rover = 89.56 - 440.4 +467.5 = 116.66 W
- In the Cool Case:
    - Q_net out = Q_net in + Q_gen_net
    - Q_rad_rover + Q_cond1 + Q_cond2 = Q_gen_rover
    - Qcond1+2 = 73.97 W
    - Q_rad_rover = 26.9 W
    - Q_gen_rover = 73.97+ 26.9 = 100.87 W

Thermal Spreadsheet

Cold Case	Tsurface = 27 K	T rover = 193 K					
Conduction	k	A	L	T1	T2		
Wheel = 12.32 W	17.4	0.00175	0.41	193	27		
Qcond1+2	12.32*6	Qcond1+2 = 73.97 W					
Radiation	e	sigma	F	A	T1	T2	
Q_rover	1	5.67E-08	1	0.342	193	27	
							Q_rover = 26.9 W
Q_gen_rover	100.87 W						
Hot Case	Tsurface = 394 K	T rover = 193 K					
Conduction	k	A	L	T1	T2		
Wheel	17.4	0.00175	0.41	193	394	Q_wheel = 14.93	
Qcond1+2 =	89.57 W						
Radiation	e	sigma	F	A	T1	T2	
Q_rover	1	5.67E-08	1	0.342	193	394	
							Q_rover = 440.4 W
Q_sun (flux)	qsolar = 1367	A = 0.342	Q_sun = 467.5 W				
Q_coolers_rover	116.66 W						

Gantt Chart (1/2)

ID#	TASK	ASSIGNED TO	PROGRESS	START	END	DAYS	MARGIN
1	**Final Design Post PDR**		0%	10/1/24	5/9/25	221	22
1.8	Schedule Margin			4/17/25	5/8/25	22	3
1.9	◆Full-scale CAD assembly of rover subsystems		Not complete	5/9/25	5/9/25	1	
2	**Ordering and Procurement**		0%	5/12/25	7/16/25	66	14
2.5	Order NSS and NIRVSS	Science	Not complete	5/12/25	7/1/25	51	6
2.6	Verify Test Facility Availability	Programmatics	Not complete	6/9/25	7/1/25	23	3
2.7	Schedule Margin			7/2/25	7/15/25	14	2
2.8	◆ Completion of Initial Component Orders		Not complete	7/16/25	7/16/25	1	1
3	**Manufacturing and Fabrication**		0%	7/17/25	12/19/25	156	14
3.3	Preliminary tests for integration	All	Not complete	11/14/25	12/5/25	22	3
3.4	Schedule Margin			12/5/25	12/18/25	14	2
3.5	◆ Completion of Fabrication for Test		Not complete	12/18/25	12/19/25	2	1
4	**Integration and Assembly**		0%	12/22/25	8/28/26	250	22
4.1	Assembly of each Subsystem	All	Not complete	12/22/25	6/30/26	191	20
4.2	Integrating subsystems into test rover	All	Not complete	6/30/26	8/5/26	37	4
4.3	Schedule Margin			8/6/26	8/27/26	22	3
4.4	◆ Systems Integration Review (SIR)		Not complete	8/28/26	8/28/26	1	1
4.5	◆ Critical Design Review (CDR)			9/30/26	9/30/26	1	1

Gantt Chart (2/2)

5	**Subsystem Testing**		0%	10/1/26	6/11/27	254	28
5.3	TVAC testing of internal components	Thermal/Mechanical	Not complete	10/1/26	5/14/27	226	23
5.8	Instrument Calibration Tests	Science/Mechanical	Not complete	10/1/26	5/14/27	226	23
5.9	Schedule Margin			5/14/27	6/10/27	28	3
5.1	◆ Subsystem Testing Complete		Not complete	6/11/27	6/11/27	1	1
6	**System-level testing**		0%	6/11/27	9/17/27	99	14
6.3	Simulated Emergency testing	All	Not complete	7/16/27	7/30/27	15	2
6.4	Mock Mission Testing	All	Not complete	7/30/27	9/3/27	36	4
6.5	Schedule Margin			9/3/27	9/16/27	14	2
6.6	◆ Integrated Testing Complete		Not complete	9/17/27	9/17/27	1	1
7	**Launch Preparation**		0%	9/20/27	12/17/27	89	15
7.1	Create final summary of risk analysis	Programmatics	Not complete	9/20/27	10/4/27	15	2
7.2	Verify mission operational status	Systems	Not complete	9/20/27	10/25/27	36	4
7.3	Prepare Launch Readiness Review	All	Not complete	9/20/27	12/3/27	75	8
7.4	Schedule Margin			12/3/27	12/17/27	15	2
7.5	◆ Launch Readiness Review (LRR)		Not complete	12/17/27	12/17/27	1	1
7.6	◆ Launch Date		Not complete	9/1/28	9/1/28	1	1

TEAM NOTES (not in presentation)

- Do not write that much per slide, show / write a little on slides, then explain